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Smart Inactive Strings Management Solution to Maximize Well Availability

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Abstract

As part of its commitment to Operational Excellence, ADNOC aims to maximize production from its extensive well stocks across Onshore and Offshore fields, which include thousands of producers and injector strings. Effective string management is crucial for production planning and operations. Implementing a standardized process for managing string stock is crucial for effective governance, ensuring the maximum availability of healthy producing strings. However, monitoring strings across multiple fields poses significant challenges. This paper introduces the implementation of a Smart Inactive String Management Solution, which offers analytical insights to engineers, enabling them to effectively manage inventories, maximize well uptime, and enhance production.

The digital solution was developed in accordance with company-wide guidelines to standardize inactive categories across ADNOC's fields. It seamlessly integrates with the operations database, offering comprehensive visibility of inactive strings along with critical monitoring KPIs. Smart analytics provide value-oriented insights by analyzing historical string statuses alongside production and injection statistics. String up time and downtime analysis assesses production impact across various scenarios. Dynamic visualizations empower users to identify the root causes of inactivity, enabling timely interventions through historical analysis and status change tracking.

The tool's analytical capabilities enable stakeholders to assess inactive categories and their business impact effectively. Its implementation has helped to reduce inactive strings through proactive management, thereby minimizing downtime and maximizing availability. The interactive UI provides critical KPIs at users' fingertips, allowing for the rapid identification of string failure reasons and expedited reactivation. Secure data management ensures appropriate accessibility in line with asset-specific protocols.

Monthly inactive string KPIs offer a clear assessment of the impact across various categories, while problem tracking prioritizes high-value reactivation opportunities. Smart analytical insights into failure mechanisms, such as high WC, high GOR, reservoir issues, and facility constraints, have improved well availability. Tracking activities that contributed to production gains has generated valuable operational knowledge.

The solution integrates with AI Production System Optimization (AiPSO), ADNOC's Digital Twin of Production System which provides real-time surveillance capabilities, model and data driven engineering

workflows to optimize production. AiPSO enables the validation of the string status and automatically identify reactivation opportunities, enhancing the inactive string assurance process.

Smart analytics combined with interactive visualization facilitate focused decision-making on string status, optimizing well movements and value realization. This advanced well-health assessment offers essential components for smart surveillance, reducing inactive durations and production losses. Future plans include incorporating AI/ML and data-driven methods to further enhance string management.

Introduction

In alignment with ADNOC's strategic digital transformation agenda, enhancing well availability has become a pivotal performance metric across its asset portfolio. The management of thousands of well strings—each subject to unique operational, reservoir, and facility constraints—demands a unified and intelligent approach to surveillance and categorization.

Historically, the absence of standardized definitions and fragmented data sources has impeded effective governance and timely decision-making. To address these challenges, ADNOC has prioritized the development of a robust framework for inactive string management, aimed at improving asset integrity, optimizing production, and driving operational excellence.

This paper introduces the Smart Inactive String Management Solution, a data-driven platform designed to streamline string surveillance across ADNOC's Onshore and Offshore fields. The solution integrates corporate guidelines, advanced analytics, and standardized workflows to enhance visibility, governance, and decision-making. By enabling consistent categorization and proactive intervention, the solution supports ADNOC's broader objectives of maximizing well uptime, minimizing deferred production, and fostering digital excellence across its operations.

Business Challenges

Inactive strings present a significant constraint in achieving optimal well availability across ADNOC's asset portfolio. These strings, if not addressed promptly, contribute to deferred production, reduced asset efficiency, and elevated operational risk. The absence of standardized categorization and inconsistent surveillance practices across any field further exacerbates the challenge, often resulting in delayed interventions and missed value opportunities.

Key challenges include:

- **Non-Uniform Categorization:** Variability in the definition and classification of inactivity across assets impedes cross-field benchmarking and governance.
- **Fragmented Data Ecosystems:** Surveillance data is dispersed across multiple platforms, limiting end-to-end visibility and hindering integrated analysis.
- **Limited Diagnostic Capability:** Require smart analytical capabilities to trace inactivity to specific root causes, such as integrity failures, reservoir constraints, or flow assurance issues.
- **Delayed Reactivation Cycles:** Without a structured prioritization framework, high-value strings potentially remain inactive longer than necessary, impacting production targets.

These challenges underscore the need for a centralized, intelligent solution that integrates data, standardizes definitions, and enables advanced analytics to drive timely and informed decision-making.

Business Value

The Smart Inactive String Management Solution delivers measurable business impact by enabling proactive governance, standardized categorization, and data-driven decision-making across ADNOC's assets. By aligning technical workflows with strategic objectives, the solution supports ADNOC's four pillars: **efficiency, profitability, performance, and people**.

The core objective of the Inactive Strings Guidelines is to ensure full utilization of well stock—both producers and injectors—by minimizing downtime and restoring string activity within the shortest feasible timeframe. The prioritization framework for reactivation is based on:

- **Criticality**: Addressing integrity and HSE-related issues with immediate attention.
- **Production Gain vs. Cost**: Evaluating payback time and economic feasibility for each candidate string.
- **Strategic Production Targets**: Aligning reactivation efforts with reservoir and field-level business plans.

The solution delivers value across multiple dimensions:

- **Enhanced Governance**: Standardized inactive categories enable consistent reporting, performance tracking, and cross-asset benchmarking.
- **Improved Uptime**: Analytical insights facilitate early identification of reactivation opportunities, reducing deferred production.
- **Operational Efficiency**: Engineers are equipped with inactive KPIs and diagnostic tools to assess string health and initiate corrective actions.
- **Strategic Planning**: Historical analytics inform production forecasting, asset development, and investment prioritization.

By embedding smart analytics into operational workflows, the solution transforms inactive string management from a reactive process into a proactive, value-driven discipline supporting ADNOC's broader goals of digital excellence and sustainable production optimization.

Inactive String Management Framework

The development of the Smart Inactive String Management Solution followed a structured, collaborative, and user-centric approach, leveraging ADNOC's corporate data infrastructure and operational guidelines. The framework is designed to ensure technical robustness, operational relevance, and scalability across ADNOC's diverse assets covering People, process and technology. Figure 1 below illustrates various components of the framework.

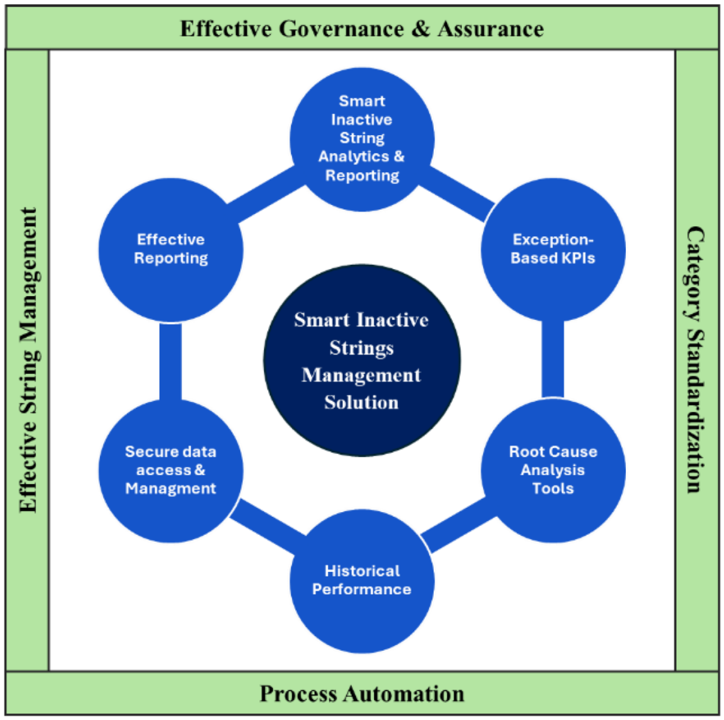

Figure 1—Inactive String Management Framework

The following core components of the framework enable effective surveillance and governance across ADNOC's assets:

- **Smart Inactive String Analytics & Reporting:** Interactive solution interface presents analytics on string status, performance metrics, and categorization trends. These visualizations empower engineers and planners to monitor well health, identify reactivation opportunities, and assess business impact with clarity and precision.
- **Exception-Based KPIs:** The system highlights strings with prolonged inactivity, significant production impact, or frequent status changes. Exception-based indicators allow users to focus on high-priority issues, enabling timely interventions and minimizing deferred production.
- **Root Cause Analysis Tools:** Visual workflows trace inactivity trends across categories and problem types, enabling users to drill down into specific assets, fields, or reservoirs. These tools support diagnostic assessments and facilitate targeted action planning.
- **Historical Performance:** Time-series visualizations provide insights into uptime and downtime patterns, category transitions, and intervention outcomes. Historical trends support performance benchmarking and strategic planning across assets.
- **Secure Data Access & Management:** Role-based access controls ensure that users view only data relevant to their assigned assets and responsibilities. Integration with ADNOC's corporate user directory enforces compliance with governance protocols and supports secure data management.
- **Effective Reporting:** Standardized reporting templates and automated data refresh cycles ensure consistency and reliability in performance tracking. These reports support cross-asset comparisons, strategic reviews, and executive decision-making.

The solution was built upon ADNOC's standardized guidelines for inactive string categorization, enabling assets to systematically define inactivity causes and implement targeted reactivation strategies. These guidelines provided a foundation for consistent classification, root cause identification, and timely intervention planning.

Key elements of the solution development process included:

- **Stakeholder Engagement**: Continuous collaboration with multidisciplinary teams—Petroleum Engineering, Reservoir Management, Production Operations, Maintenance, and Planning—ensured alignment with field-specific requirements and operational realities.
- **Standardization:** Standardized string categorization streamlining definitions, reporting, and effective cross-asset benchmarking
- **Iterative Development Cycles**: Agile development enabled rapid prototyping, feedback incorporation, and refinement of analytical features and visualizations.
- **Data-Driven Design**: Historical string performance data, operational KPIs, and production metrics informed the analytical framework, ensuring relevance and accuracy.
- **Scalable Architecture**: The solution was designed for deployment across multiple assets with minimal customization, supporting ADNOC's enterprise-wide digital strategy.

This structured solution development process ensured consistent categorization, facilitating root cause analysis, and supporting timely reactivation planning—enabling ADNOC to optimize well availability and production efficiency across its portfolio.

Categorization

Central to the solution is a comprehensive string categorization model that enables accurate status identification and root cause allocation. This model supports proactive planning and execution of remedial actions across assets.

The categorization includes:

- **Total Strings**: Encompasses all company strings, segmented into Abandoned and Operational categories.
- **Operational Strings**: Further classified into Suspended, Approved P&A, Standby, Projects, Active, and Inactive strings.

Each sub-category is defined as follows:

- **Suspended Strings**: Temporarily out of service but eligible for reactivation based on operational plans.
- **Active Strings**: Currently contributing to production or injection, including both healthy and problematic strings under surveillance.
- **Standby Strings**: Capable of flow but shut-in due to quota limits, market conditions, or strategic planning.
- **Projects**: Strings linked to ongoing development or EPC tie-in activities.
- **Approved P&A**: Strings designated for permanent abandonment.
- **Inactive Strings**: Strings unable to produce or inject due to surface or subsurface issues requiring intervention. These are further classified into 5 main categories:
  - **Integrity-Related Issues:** Strings affected by wellhead, subsurface, or facility integrity concerns.
  - **Tie-In Delays:** Newly drilled or worked-over strings awaiting hook-up or commissioning.
  - **Flow Assurance Issues:** Strings impacted by scale, asphaltene, equipment failure, or artificial lift challenges.
  - **Reservoir-Related Issues:** Strings shut-in due to reservoir management guidelines, integrity concerns, or pending development activities.

- Reservoir Integrity issues (e.g. formation collapse, poor reservoir quality)
- Reservoir Management Guidelines (e.g. high GOR issues, high water cut, low reservoir pressure, low flowing bottom hole pressure, low reservoir productivity, low wellhead pressure, etc.)
- Strings that are unable to flow due to reservoir problems that could be addressed with better artificial lift or better completion interface, i.e. high water cut, dead wells etc.

- **Dynamic Strings:** Temporarily inactive due to operational constraints such as nearby activities, facility shutdowns, or HSE limitations.

The Figure 2 below sub-categorizes all company well strings with emphasis on Inactive Strings.

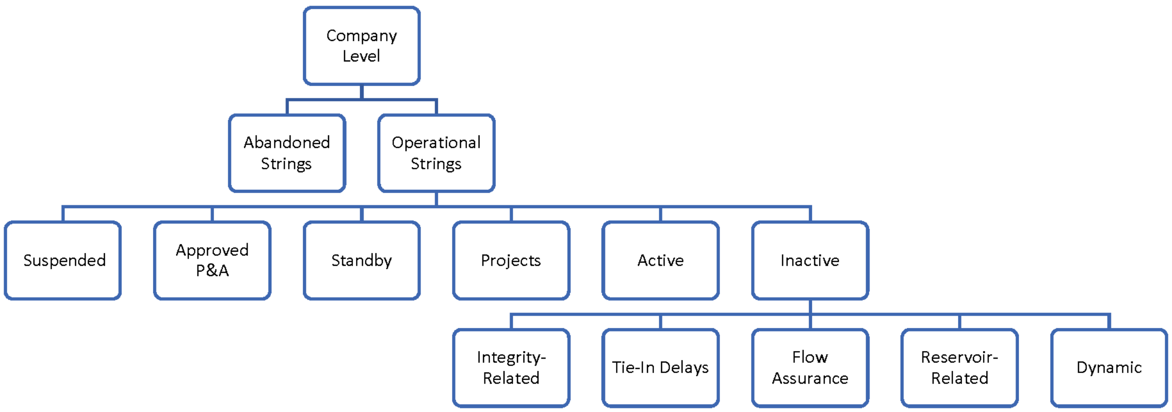

Figure 2—Inactive String Categories

Standardization

To ensure consistency, comparability, and governance across ADNOC's diverse operational landscape, the Smart Inactive String Management Solution incorporated a standardized taxonomy for string categorization. This corporate-level standardization effort harmonized definitions, streamlines reporting, and enabled effective cross-asset benchmarking.

Standardization has been instrumental in:

- **Enhancing Governance**: Unified definitions across fields support consistent performance tracking and reporting.
- **Facilitating Root Cause Analysis**: Categorization aligned with operational, reservoir, and facility domains enables targeted diagnostics.
- **Driving Uniformity**: Standardized workflows and KPIs ensure consistent surveillance and intervention planning across assets.

The categorization framework spans multiple dimensions critical to well health and availability, including:

- Reservoir and Sector
- Well Name and Status
- Inactive Category
- Field and Asset Identifiers

Operational string categories were defined and standardized across ADNOC group companies, including:

- Active Strings
- Inactive Strings
- Operational Strings
- Standby Strings
- Disposal Strings
- Suspended Strings
- Abandoned Strings
- Problematic Active Strings
- Healthy Strings
- Plug and Abandonment (P&A) Strings
- Flow Assurance
- Simultaneous Operations
- Workover Wells
- Subsurface Reservoir
- Well Integrity

In addition to categorization, the guidelines included structured action plans, problem types, and performance KPIs for each asset and team. These are monitored and updated at regular intervals to ensure alignment with operational goals.

The solution embedded these standards into its analytics and visualization layers, ensuring that every insight adheres to ADNOC's established guidelines. This enabled end users to assess string availability, status, and operational impact with clarity and confidence—empowering informed decision-making and proactive management across all assets.

Inactive String Management Solution Architecture

The Smart Inactive String Management Solution is built on a scalable, modular architecture designed to enable effective String management through standardization, advanced analytics, and secure data governance across ADNOC's upstream assets. The architecture integrates seamlessly with ADNOC's corporate data infrastructure, enabling consistent performance, reliability, and extensibility. Below Figure 3 illustrates core components of the solution architecture.

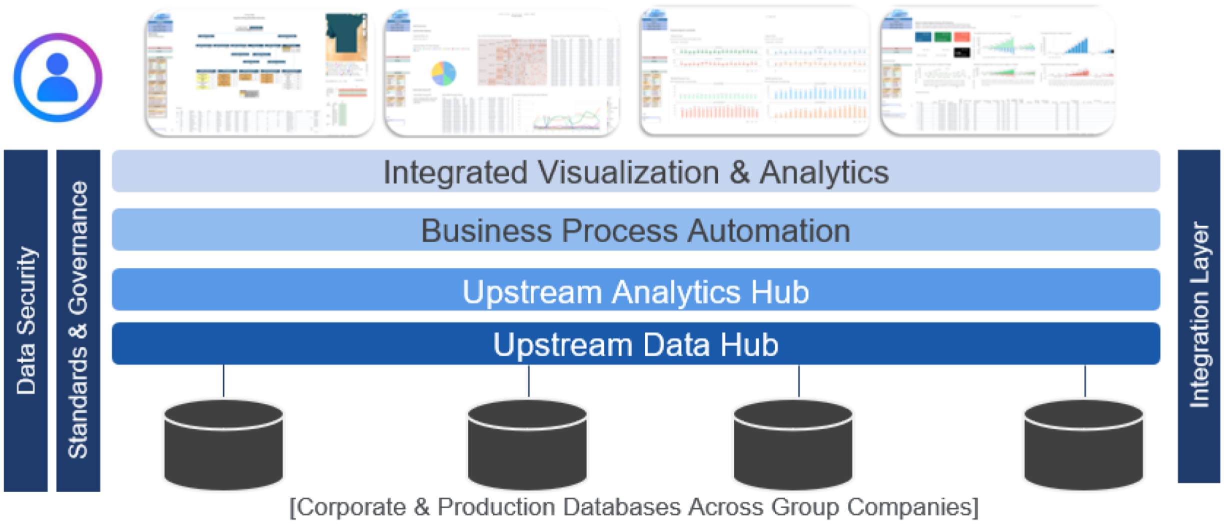

Figure 3—Inactive String Management Solution Architecture

Core Components

- **Data Ingestion Layer**: Automated ETL (Extract, Transform, Load) pipelines extract and transform data from multiple operational databases, ensuring timely updates and standardized input across all assets.
- **Analytics Engine**: A centralized analytical core processes historical and current data to generate KPIs, categorize string statuses, and deliver key insights.
- **Visualization Layer**: Role-based components provide intuitive access to performance metrics, category trends, and reactivation opportunities. Embedded visual workflows support strategic planning and operational decision-making.
- **Integration Layer**: The solution is integrated with ADNOC's AiPSO platform, enabling model-driven validation of string status and identification of high-value reactivation candidates through digital twin capabilities.
- **Security & Access Control**: Role-based access is enforced through integration with ADNOC's corporate user directory, ensuring compliance with RACI protocols and safeguarding sensitive operational data.

Data Architecture & Management

The Smart Inactive String Management Solution is underpinned by a robust and scalable data architecture that integrates diverse datasets across ADNOC's operational landscape. The architecture is designed to ensure data integrity, accessibility, and performance while supporting advanced analytics and operations decision-making. The solution seamlessly integrates with ADNOC's corporate data hubs, enabling seamless connectivity to field-level data sources. A standardized data pipeline was established to extract, transform, and load information into the analytics engine, ensuring consistency and accuracy across all assets.

The solution ingests and processes both static and current datasets from multiple sources, enabling comprehensive surveillance and analysis of string performance. Key architectural components include:

- **Standardized Data Models**: Unified schemas ensure consistent interpretation of string status, performance metrics, and categorization across fields.
- **Upstream Data Hub**: A centralized platform that aggregates and standardizes operational and reservoir data from ADNOC's group companies, enabling unified access and consistent

surveillance. It ensures data integrity and harmonization across assets, forming the foundation for analytics and governance.

- **Upstream Analytics Hub**: An advanced analytics engine that leverages advanced technological features to generate insights, categorize string statuses, and support predictive decision-making. It transforms raw data into actionable intelligence, enhancing well performance and optimizing production strategies.
- **Data Ingestion Pipelines**: Automated ETL (Extract, Transform, Load) processes continuously pull data from operational databases across assets.
- **Transformation Layers**: Business logic is applied to standardize string categorization, compute KPIs, and prepare data for analytics.
- **Storage Mechanisms**: A scalable centralized infrastructure ensures high availability, reliability, and performance across all assets.

Key features of data management include:

- **Automated Updates**: ETL workflows are configured to reflect the latest data inputs, minimizing latency and ensuring most recent insights.
- **Data Quality Assurance**: Pre-processing routines validate and cleanse incoming data, reducing manual effort and enhancing analytical reliability.
- **Access Controls**: Role based access and entitlements is designed as per the RACI. This ensures that only authorized personnel can access asset-specific insights.

This architecture not only supports the current analytical capabilities of the solution but also provides a scalable foundation for future enhancements, including AI/ML integration and predictive analytics.

Visualization and Analytics

User experience is a central pillar of the Smart Inactive String Management Solution. The visualization layer is designed to deliver actionable insights through intuitive, role-based components that support rapid decision-making and operational responsiveness. Advanced analytics form the core of the solution's value proposition, enabling predictive insights, failure diagnostics, and strategic prioritization. The analytical models are designed to assess category performance, identify root causes, and forecast reactivation potential. Below Figure 4 illustrates various visualization and analytical features as part of the Smart Inactive String Management Solution.

Figure 4—Visualization and Analytics

Key Visualization Features

The visualization layer of the Smart Inactive String Management Solution is designed to deliver actionable insights through intuitive, role-based components that support rapid decision-making and operational responsiveness. Below Figure 5 summarizes key visualization features developed in the solution.

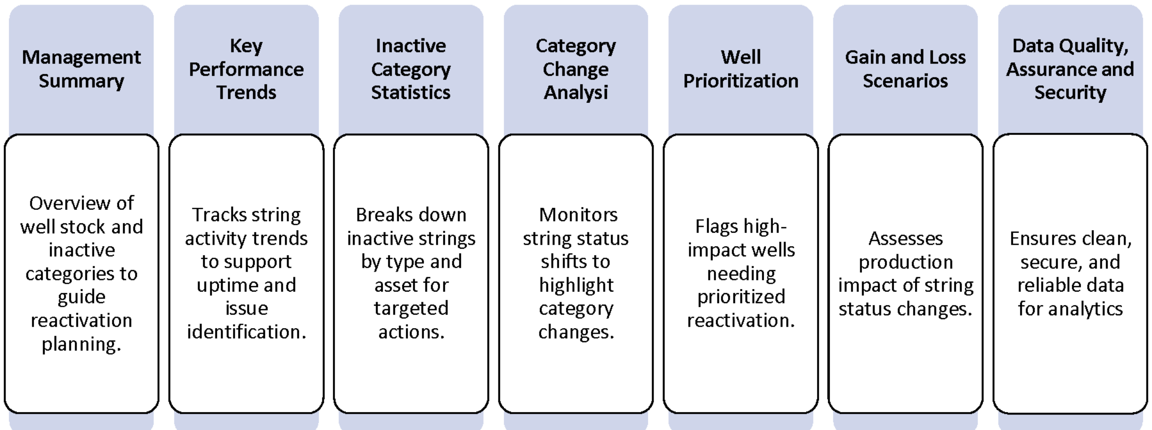

Figure 5—Key Visualization Features

Management Summary

The Management Summary provides a comprehensive overview of ADNOC's total well stock, segmented into operational categories and inactive classifications. This hierarchical view enables stakeholders to assess string availability, monitor performance, and identify high-impact opportunities for reactivation.

Exception-based reporting highlights strings across various categories, problem types, and inactivity durations. Integration with last known production and injection data enhances visibility into deferred volumes and potential value realization. This structured insight supports strategic planning in order to minimize downtime and improve production efficiency.

The summary also facilitates cross-asset comparisons, enabling teams to benchmark performance and prioritize interventions based on business impact. Below Figure 6 shows an overview section of management summary.

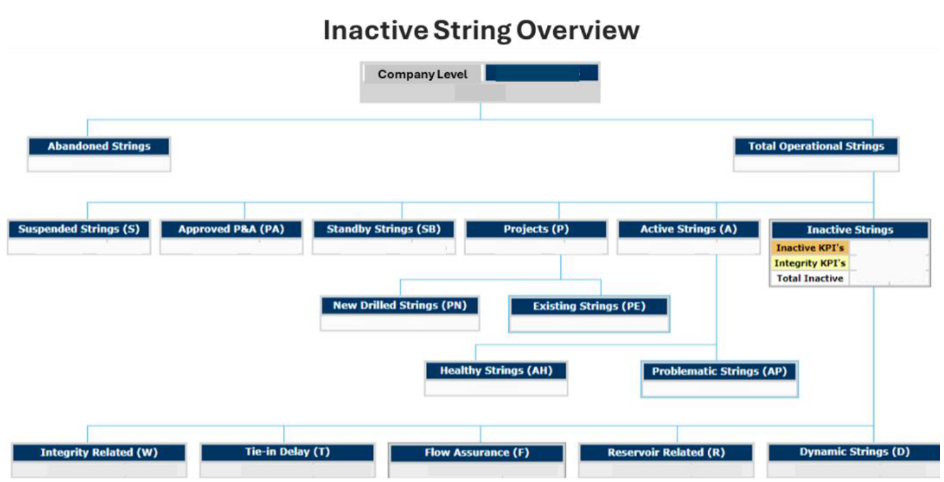

Figure 6—Inactive String Overview

Key Performance trends

This section presents analytical trends that reflect the performance of string categories over time. Historical and current data are visualized to support surveillance, planning, and optimization efforts.

Key trends include:

- **Active vs. Inactive Strings**: Comparative analysis of total active and inactive strings across assets, highlighting uptime optimization opportunities.
- **Integrity-Related Inactive Strings**: Trends in strings impacted by integrity issues, supporting proactive integrity management.
- **Reservoir-Related Inactive Strings**: Insights into strings affected by reservoir constraints, including high GOR, high water cut, and low-pressure scenarios.
- **Flow Assurance Inactive Strings**: Monitoring of strings impacted by surface and subsurface flow assurance problems.
- **Tie-In and Dynamic Strings**: Tracking of strings delayed due to commissioning or operational constraints.

Historical categorization trends, based on standardized definitions, provide users with a clear understanding of performance evolution. Month-over-month comparisons enable field teams to assess immediate changes, identify emerging issues, and refine surveillance strategies.

These performance insights empower ADNOC teams to focus on high-value string groups, optimize well movements, and enhance overall asset productivity.

Inactive category statistics

This section provides a detailed statistical breakdown of inactive strings across ADNOC's assets, enabling users to monitor performance, identify trends, and prioritize interventions. The statistics are presented both for the current reporting period and across historical timeframes, offering a comprehensive view of string availability and uptime optimization.

Key features include:

- **Category-Level Insights**: Counts and relative proportions of strings across inactive categories such as integrity issues, reservoir related issues, flow assurance problems, tie-in delays, and dynamic operational limitations.
- **Asset and Field-Level Breakdown**: Statistics are segmented by well, reservoir, asset, field, and well type, supporting granular analysis and targeted planning.
- **Historical Comparisons**: Time-series data enables users to benchmark current performance against previous months, highlighting improvements or emerging challenges.
- **Well Type Classification**: Strings are categorized by types such as oil and gas producers and injectors, CO_2 injectors, disposal wells, observers, suspended wells, and water supply wells—providing clarity on operational impact and prioritization.

These insights establish a performance baseline and guide surveillance efforts, ensuring that teams remain focused on high-impact string groups and maintain alignment with ADNOC's production and injection objectives.

Category change analysis

The Category Change Analysis module provides a dynamic view of string status transitions, enabling stakeholders to monitor operational shifts and assess their impact on well availability. By tracking strings that move between categories—such as from active to inactive due to reservoir, integrity, or flow assurance issues, the solution offers actionable insights into emerging challenges and intervention needs.

This analytical layer highlights both immediate and historical trends, supporting performance benchmarking and strategic planning. An increasing trend in category changes may indicate rising operational constraints, while a declining trend reflects improved stability and uptime. Exception-based reporting identifies strings undergoing frequent status changes, flagging them for proactive review and early reactivation planning.

By visualizing these transitions, the solution empowers teams to anticipate downtime risks, optimize surveillance efforts, and maintain alignment with production and injection targets—transforming reactive monitoring into predictive asset management.

- Active to Inactive
  - Active to reservoir related issues
  - Active to flow assurance issue
  - Active to Tie-in delays
  - Active to integrity issue
  - Active to dynamic strings
- Inactive to active strings
  - Reservoir related issues to active
  - Flow Assurance Issue to active
  - Tie-in delays to active
  - Integrity issue to active
  - Dynamic strings to active
- Active to other string type (abandoned, suspended, standby etc.)
- Other string type (suspended, standby, project etc.) to active

Well prioritization

The well prioritization module within the Smart Inactive String Management Solution provides targeted insights into high-impact wells that offer significant production or injection potential upon reactivation. Leveraging exception-based analytics, the system identifies top-priority strings by correlating problem types, planned action dates, and operational constraints. This enables engineers to focus on wells with the highest opportunity value, ensuring timely interventions aligned with business objectives. The prioritization module supports strategic planning by highlighting wells that require immediate attention, thereby accelerating value realization and enhancing overall asset performance.

Gain Loss value analysis scenario

Advanced analytics within the solution evaluate production and injection data in conjunction with string status, type, and inactive categorization to quantify business impact. The gain-loss scenario analysis models the effect of string transitions—such as activation or deactivation—on production and injection targets over defined timeframes. These insights allow stakeholders to assess both immediate and cumulative impacts of string availability on field performance. By identifying strings that contribute positively upon reactivation, the solution supports informed decision-making and strategic optimization of well stock.

Data quality, assurance and security

Robust data quality and governance are foundational to the effectiveness of the Smart Inactive String Management Solution. The platform employs advanced pre-processing techniques to standardize and cleanse data across ADNOC's diverse asset landscape, ensuring analytical reliability and consistency. Automated ETL workflows facilitate current updates, minimizing latency and enabling timely insights. Integrated with ADNOC's corporate user directories, the solution enforces role-based access controls, ensuring secure and compliant data usage aligned with the RACI framework. Additionally, the platform provides a structured environment for data assurance teams to implement best practices, optimize well surveillance strategies, and uphold operational integrity across all assets.

Value realization

The solution's advanced analytics empower users to quantify the impact of inactive strings and identify high-value opportunities for reactivation. Key metrics include:

- String Availability and Status
- Production and Injection Losses
- Uptime and Downtime Trends
- Total Well Stock per Asset
- Deferred Volume Impact
- Gain-Loss Scenarios
- Inactive Category and Problem Type Analysis

By linking string performance with business objectives, the solution enables field teams to convert surveillance insights into actionable strategies. This proactive approach minimizes downtime, enhances availability, and drives value realization across ADNOC's operations.

Conclusion and Way Forward

The Smart Inactive String Management Solution has driven transformational improvements in well availability, operational efficiency, and governance across ADNOC's upstream assets. Central to its success is the implementation of effective inactive string governance, which ensures consistent oversight, accountability, and strategic alignment across operations.

By standardizing inactive string categorization, integrating advanced analytics, and enabling proactive intervention, the solution directly addresses critical challenges in string surveillance and production optimization. This structured approach enhances visibility, reduces ambiguity, and fosters timely decision-making.

The platform has empowered multidisciplinary teams to make informed, data-driven decisions, significantly reducing inactive durations and boosting asset performance. Through continuous information harnessing, the solution lays a robust foundation for evolving AI and data driven capabilities, enabling predictive insights and intelligent automation.

This initiative reflects ADNOC's unwavering commitment to operational excellence, digital transformation, and the strategic use of technology to optimize production and maximize value across its asset portfolio.

Through continuous information harnessing, the solution lays a robust foundation for evolving AI and data driven capabilities, enabling predictive insights and intelligent automation. We aim to fully integrate the solution into ADNOCs AiPSO intelligent surveillance and optimization platform supporting ADNOC's vision for digital excellence, operational resilience, and sustainable production growth.

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